~~Step 1~~

1. Computing predicted Values.

Given that: $x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Matrix multiplication:

$$\hat{y} = \begin{bmatrix} 1(-1) + 3(2) \\ 4(-1) + 10(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Prediction error:

$e = \hat{y} - y$ $\quad e = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

STEP 2: GRADIENT OF THE MSE cost FUNCTION

Given:

$$X=\begin{bmatrix}1 & 8\\ 4 & 10\end{bmatrix},\ y=\begin{bmatrix}5\\ 6\end{bmatrix},\ m=\begin{bmatrix}-1\\ 2\end{bmatrix},\ b=\begin{bmatrix}1\\ 1\end{bmatrix},\ n=2$$

$$\hat{y}=Xm+b$$

prediction error:

$$e=\hat{y}-y$$

cost function (MSE):

$$J(m,b)=\frac{1}{n}e^Te=\frac{1}{n}\begin{bmatrix}e_1 & e_2\end{bmatrix}\begin{bmatrix}e_1\\ e_2\end{bmatrix}=\frac{1}{n}\left(e_1^2+e_2^2\right)$$

chain of dependence:

$$J\rightarrow e\rightarrow \hat{y}\rightarrow (m,b)$$

Derivative of J with respect to $\hat{y}$ (where I'll do both gradients)

$$\frac{\partial J}{\partial e_i}=\frac{2}{n}e_i \Rightarrow \frac{\partial J}{\partial e}=\frac{2}{n}\begin{bmatrix}e_1\\ e_2\end{bmatrix}=\frac{2}{n}e$$

$$\frac{\partial e}{\partial y}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$$

$$\frac{\partial J}{\partial \hat{y}}=\left(\frac{\partial e}{\partial \hat{y}}\right)^T\frac{\partial J}{\partial e}=I\cdot\frac{2}{n}e \Rightarrow \frac{\partial J}{\partial \hat{y}}=\frac{2}{n}(\hat{y}-y)$$

Gradient with respect to m:

$$\frac{\partial \hat{y}}{\partial m} = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} = X$$

$$\frac{\partial J}{\partial m} = \left(\frac{\partial \hat{y}}{\partial m}\right)^T \frac{\partial J}{\partial \hat{y}} = X^T \cdot \frac{2}{n}(\hat{y} - y)$$

$$\Rightarrow \frac{\partial J}{\partial m} = \frac{2}{n} X^T (\hat{y} - y)$$

Gradient with respect to b:

$$\frac{\partial \hat{y}}{\partial b} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\frac{\partial J}{\partial b} = \left(\frac{\partial \hat{y}}{\partial b}\right)^T \frac{\partial J}{\partial \hat{y}} = I^T \cdot \frac{2}{n}(\hat{y} - y)$$

$$\Rightarrow \frac{\partial J}{\partial b} = \frac{2}{n}(\hat{y} - y)$$

Evaluation at iteration 0

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix} \text{ and } e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

cost: $J = \frac{1}{2} \begin{bmatrix} 1 & 11 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \frac{1}{2}(1^2 + 11^2) = \frac{1+121}{2} = \frac{122}{2} = 61$

$\Rightarrow \underline{61}$

Gradient with respect to m:

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1(1) + 4(11) \\ 3(1) \quad 10(11) \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} 45 \\ 113 \end{bmatrix} \Rightarrow \frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Gradient with respect to b:

$$\frac{\partial J}{\partial b} = \frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \Rightarrow \frac{\partial J}{\partial b} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Gradient Descent: First iteration

Initial Values from instructions and Calculations:

$$m_0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad b_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \hat{y}_0 = \begin{bmatrix} 6 \\ 17 \end{bmatrix}, \quad e_0 = \begin{bmatrix} 1 \\ 11 \end{bmatrix},$$

$J_0 = 61$ Let $\alpha = 0.01$

update m: $m_1 = m_0 - \alpha \frac{\partial J}{\partial m}$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.45 \\ 1.13 \end{bmatrix}$$

$$m_1 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

update b: $b_1 = b_0 - \alpha \frac{\partial J}{\partial b}$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ 0.11 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

Recalculate predictions:

$$\hat{y} = Xm_1 + b_1$$

$$\hat{y}_1 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\hat{y}_1 = \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} //$$

recalculate errors:

$$e_1 = \hat{y}_1 - y = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} //$$

New MSE:

$$J_1 = \frac{1}{2} e_1^T e_1$$

$$= \frac{1}{2}((-2.85)^2 + (-2.21)^2)$$

~~$= -4.8841 +$~~

~~$= -4.8841 +$~~

$$J_1 = \frac{8.1225 + 4.8841}{2} = \frac{13.0066}{2} =$$

$$= 6.5033 //$$

First Iteration Summary

$\alpha = 0.01$

$m_1 = [-1.45, 0.87]$

$b_1 = [0.99, 0.89]$

New prediction = $[2.15, 3.79] = \hat{y}$

New error (e_1) = $[-2.85, -2.21]^T$

New MSE = 6.5033

Calculate new gradients from new parameters.

Gradient with respect to m:

$$\frac{\partial J}{\partial m} = X^T e_1 = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-2.85) + 4(-2.21) \\ 3(-2.85) + 10(-2.21) \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

Gradient with respect to b.

$$\frac{\partial J}{\partial b} = 2 e_1$$

Since $n = 2$: $\frac{\partial J}{\partial b} = \frac{2}{2} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$

Gradient Descent: Update 2

Starting values: $\alpha = 0.01$, $n = 2$

$$m_1 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}, \quad \hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}, \quad J_1 = 6.5033$$

Gradients at m_1, b_1

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e_1 = \begin{bmatrix} -11.69 \\ 30.65 \end{bmatrix}, \quad \frac{\partial J}{\partial b} = \frac{2}{n} e_1 = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

Update m:

$$m_2 = m_1 - \alpha \frac{\partial J}{\partial m} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - \begin{bmatrix} -0.1169 \\ -0.3065 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$$

Update b:

$$b_2 = b_1 - \alpha \frac{\partial J}{\partial b}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - \begin{bmatrix} -0.0285 \\ 0.0221 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

Recalculate predictions:

$$\hat{y}_2 = X m_2 + b_2 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} + \begin{bmatrix} 0.0185 \\ 0.9121 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1.3331) + 3(1.1765) \\ 4(-1.3331) + 10(1.1765) \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3331 + 3.5295 \\ -5.3324 + 11.7650 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

$$= \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} = \hat{y}_2 = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$$

Recalculate errors:

$$e_2 = \hat{y}_2 - y = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

New MSE:

$$J_2 = \frac{1}{2} e_2^T e_2$$

$$= \frac{1}{2}\left((-1.7851)^2 + (1.3447)^2\right)$$

$$= \frac{3.1866 + 1.8082}{2} = \frac{4.9948}{2}$$

$$J_2 = 2.4974$$

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Update 3 (Member 3).

Starting value: $M = [-1.3331, 1.1765]^T$,
$b = [1.0185, 0.9121]^T$

$$XM = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} = \begin{bmatrix} -1.3331 + 3.5295 \\ -5.3324 + 11.7650 \end{bmatrix}$$

$$XM = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix}$$

But $\hat{y} = XM + b$. substitute the value

$$\hat{y} = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$$

predicted error $e = \hat{y} - y$.

$$e = \begin{bmatrix} 3.2149 - 5 \\ 7.3447 - 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

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$$\text{Cost}_J = \left((-1.7851)^2 + (1.3447)^2\right)/2$$

$$= (3.1866 + 1.8082)/2.$$

$$= 2.4974$$

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$= \begin{bmatrix} -1.7851 + 4(1.3447) \\ 3(-1.7851) + 10(1.3447). \end{bmatrix}$$

$$= \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = [3.5937, 8.0917] \quad \text{and} \quad \frac{\partial J}{\partial b} = e$$

$$e = [-1.7851, 1.3447]^T$$

$$M_2 = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01$$

$$= \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix}$$

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$$b_3 = \begin{bmatrix} 1.0155 \\ 0.9121 \end{bmatrix} - 0.01$$

$$= \begin{bmatrix} -1.7851 \\ 1.8447 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

After updating: $M = [-1.3690, 1.0956]^T$

$$b = [1.0364, 0.8987]^T$$

check with the new values.

$M = [-1.3690, 1.0956]^T$, $b = [1.0364, 0.8987]^T$

$\hat{y} = xm + b.$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1.3690) + 3(1.0956) \\ 4(-1.3690) + 10(1.0956) \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

$$= \begin{bmatrix} 1.9178 \\ 5.4800 \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

$$= \begin{bmatrix} 1.9178 + 1.0364 \\ 5.4200 + 0.3787 \end{bmatrix}$$

$$= \begin{bmatrix} 2.9542 \\ 6.3787 \end{bmatrix}$$

predicted error: $e = \hat{y} - y$

$$e = \begin{bmatrix} 2.9542 \\ 6.3787 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix}$$

$$J = (1/2)\, e'e = \frac{1}{2}\left[(-2.0458)^2 + (0.3787)^2\right]$$

$$= \frac{1}{2}\left[4.1858 + 0.1434\right]$$

$$= \underline{\underline{[2.1644]}}$$

Therefore the cost fell from 2.4974 to about 2.1644, so update 3 reduced the error.

Summary of results

The cost in each row is the cost computed with that row's M and b.

Stage	M	b	cost J
initial	[-1, 2]	[1, 1]	61.
After update 1	[-1·4500, 0·8700]	[0·9900, 0·8900]	6·5033.
After update 2.	[-1·3331, 1·1765]	[1·0185, 0·9121]	2·4974.
After update 3.	[-1·3690, 1·0956]	[1·0364, 0·8987]	2·1644.

Question 7:

Trend in M and b

- M_2 falls from 2 to about 1.10, and M_1 from -1 to -1.37. The first update is the biggest because the initial error was big.

  The prediction 17 was far from the target 6.

- b changes only slightly and stays close to 1. Its gradient (e) is much smaller than the gradient for M ($x^T e$), so it moves less.

- The first update exceeds because the predictions go from too high too low. That is why the signs of the errors flip in update 2. And the next updates are smaller and the values begin to settle.

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Yes, the parameters are moving in a direction that reduces the error. The cost drops from 61 to 6.50 Then 2.50, then 2.16 and it can reduce if more iterations are calculated.

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